

DIFFERENTIAL EQUATIONS

Shortcuts & Super Simple Tricks — By Thunderstudy

An equation that contains dy/dx (rate of change). If a normal equation tells position, a differential equation tells how it changes.

TRICK 1 — Identify TYPE in 2 Seconds

Most MCQs are solved just by recognizing the type. Identification = 70% of solving.

Look at Equation	Type	Action
$dy/dx = f(x)$	Variable separable	Direct integrate
$dy/dx + Py = Q$	Linear DE	Use formula
$dy/dx = f(y/x)$	Homogeneous	Put $y=vx$
y'' present	Order = 2	Just identify

TRICK 2 — Order & Degree (FREE MARKS)

- ◆ **Order:** Highest derivative.
- ◆ **Degree:** Power of the highest derivative.

Example: $(d^2y/dx^2)^3 + y = 0$. Order = 2, Degree = 3.

 **Shortcut: Ignore everything except the highest derivative.**

✓ TRICK 3 — Check Option Without Solving

Instead of solving: 1 Take option 2 Differentiate it 3 Substitute into equation.

👉 LHS = RHS? Done! Testing is faster than solving.

✓ TRICK 4 — Linear DE Formula (MEMORISE!)

If: $\frac{dy}{dx} + Py = Q$

$$IF = e^{\int P dx} \mid \text{Solution: } y(IF) = \int Q(IF) dx$$

MCQ Hack: Just identify P and IF to match the option pattern!

✓ TRICK 5, 6 & 7 — Fast Logic

Variable Separable: If $\frac{dy}{dx} = xy$, write $\frac{dy}{y} = x dx$ instantly.

General vs Particular: Contains '+ C'? General. Missing 'C'? Particular.

Constant Solution: If $\frac{dy}{dx} = f(y)$, set $f(y)=0$ and solve for y directly.

✓ TRICK 8 — Homogeneous SPEED RULE

If every term has the same degree (e.g., $\frac{dy}{dx} = \frac{(x+y)}{x}$), immediately put $y = vx$. MCQs often ask for the method only!

🧠 5-SECOND MCQ STRATEGY (Exam Flow)

1. Find Type (1 sec)
2. If options given → Test solution
3. Look for formula pattern (IF)
4. Avoid full integration unless forced

GOLDEN EXAM HACK (Topper Trick)

"60–70% of DE questions are Recognition (Order, Degree, Type).

Try recognition first, solving later!"

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